Which claims can the supplied evidence justify?

A tractable research question

Does aligning each training claim with the evidence actually supplied reduce unsupported assertions without losing useful diagnostic performance?

Central idea: supported. Dominant root cause: unproven.
Candidate contribution: source provenance + matched supervision intervention.

Separate four problems before choosing a method

Missing evidence

A biopsy result or prior study is absent.

Missed perception

The relevant finding is present but missed.

Ambiguous evidence

The supplied image cannot resolve the diagnosis.

Unused evidence

A retained finding loses to a text shortcut.

Teacher knowledge can exceed student evidence

Article / teacher

Images + captions
History, pathology, other panels / dates

Training record

Selected image(s)
+ generated question
+ generated answer

Possible mismatch

Answer contains a fact whose evidence never reaches the student

This is a risk to measure, not proof that all synthetic supervision is defective.

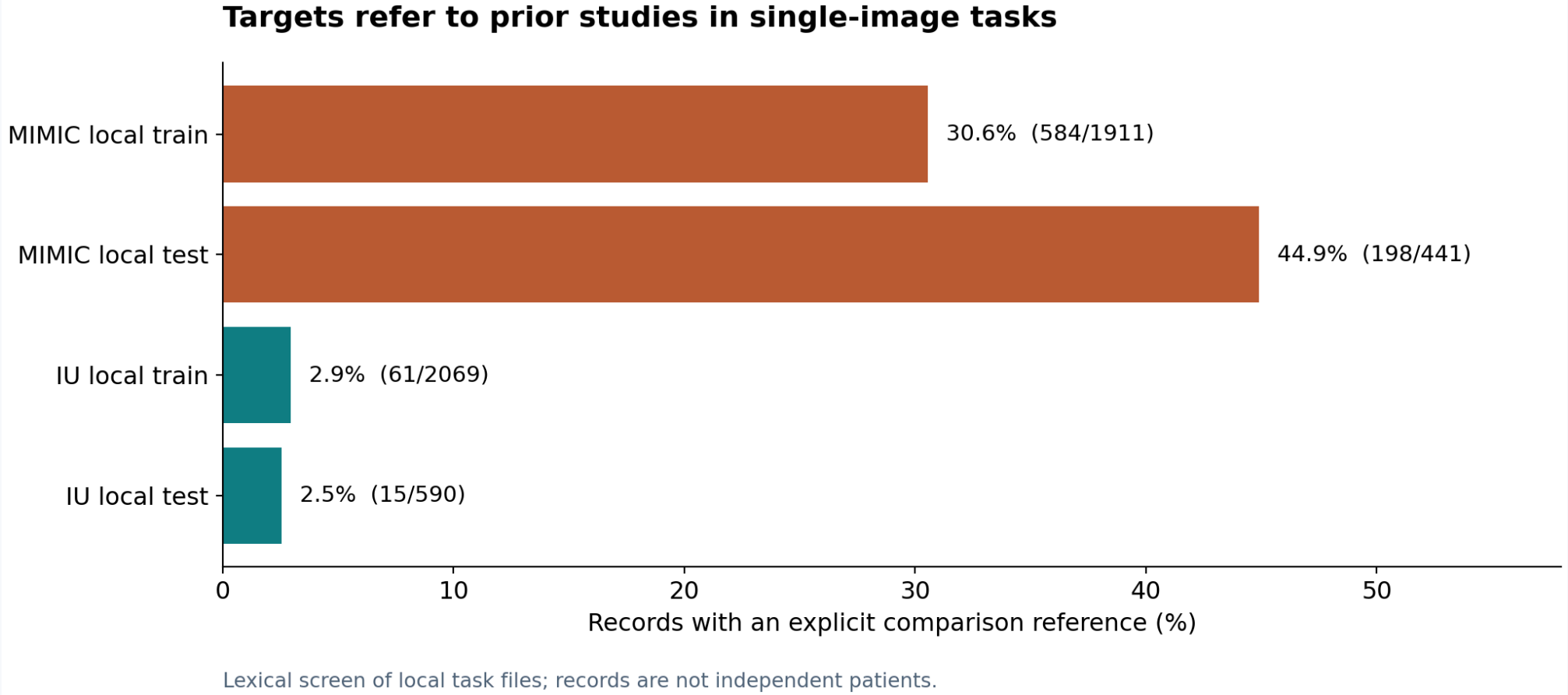
Accuracy can rise without stronger visual dependence

- MedVLM-R1: 55.11% → 78.22% in its evaluation, using 600 training QA.
- MedVLThinker: curated text-only RL can outperform its tested multimodal training recipe.
- Beyond Accuracy: accuracy 56.5% → 58.8%, while image sensitivity falls 48.2% → 39.8%.
- Test answer correctness and evidence dependence separately. Neither alone certifies a visual premise.

The latest work narrows the novelty claim

- MCR and Med-OPD: image-conditioned rewards and token weighting already exist.
- MedTRACE: evidence sufficiency, tools, verification and confidence already exist.
- CARE-MVLM: lesion removal + matched random removal + abstention already exist.
- Candidate extension: teacher-only patient facts and missing panels, tested through a controlled target-alignment intervention.

Single-image report targets refer to unavailable priors



Lexical screen of local task files; records are not independent patients.

A traceable report example

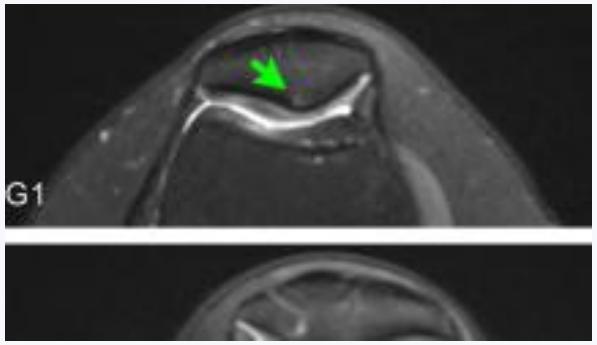
Input: one current image + “generate a concise medical report.”

Target: “As compared to the previous radiograph, no relevant change has occurred.”

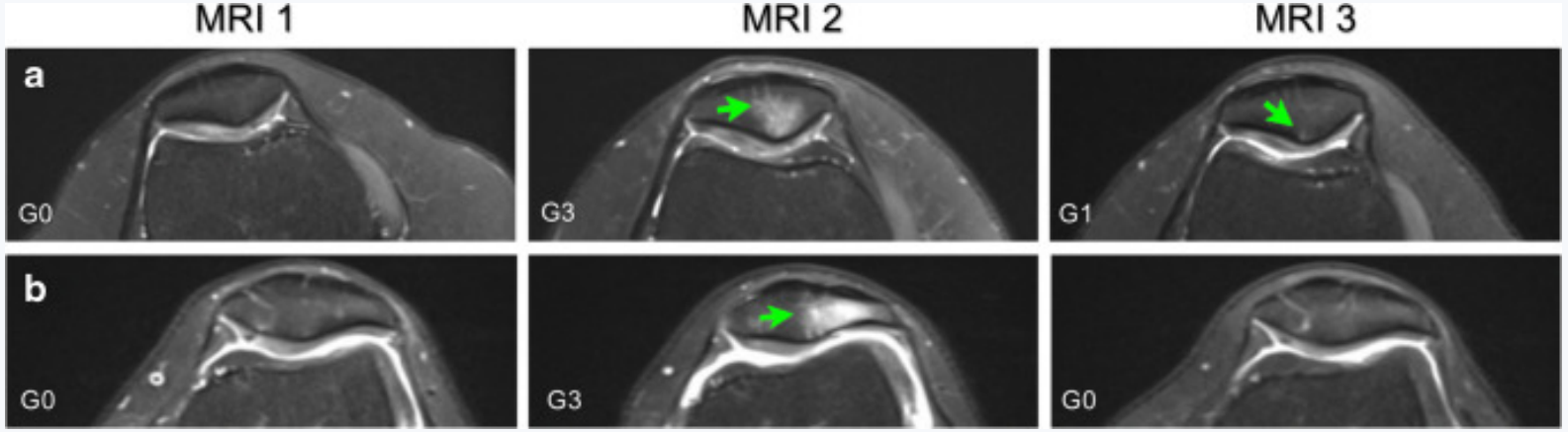
Current findings can remain valid.
The comparison needs another study. Supply it or revise the target.

A stronger case: earlier timepoints were cropped away

Actual input

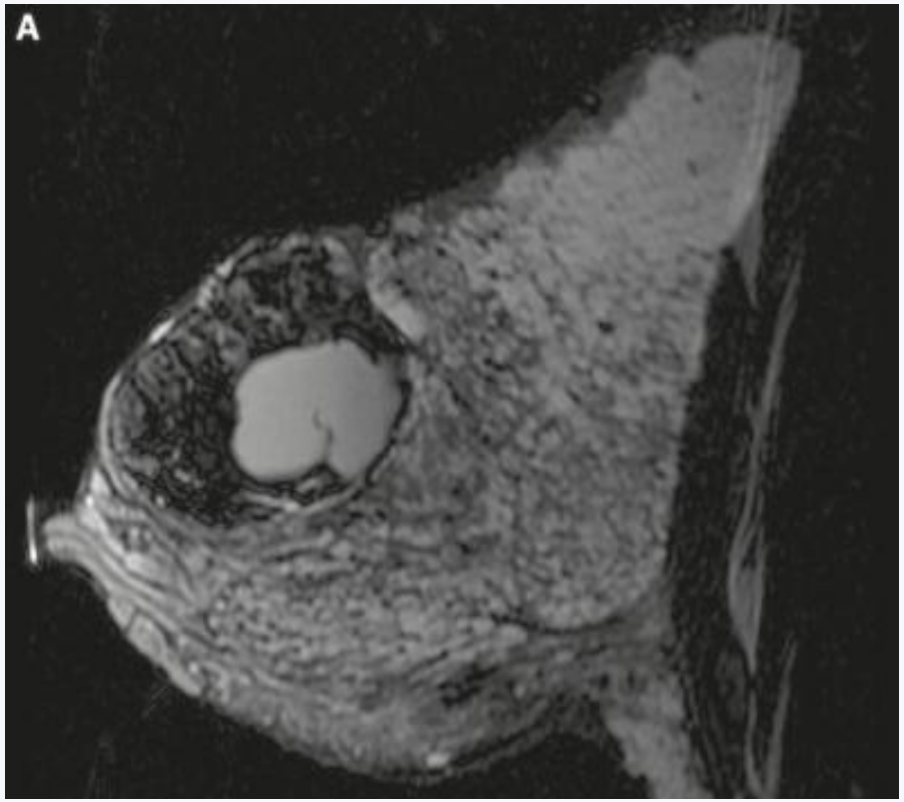


Published figure



Target describes recovery across time. Input contains only the final timepoint.

A true diagnosis can have unsupported provenance

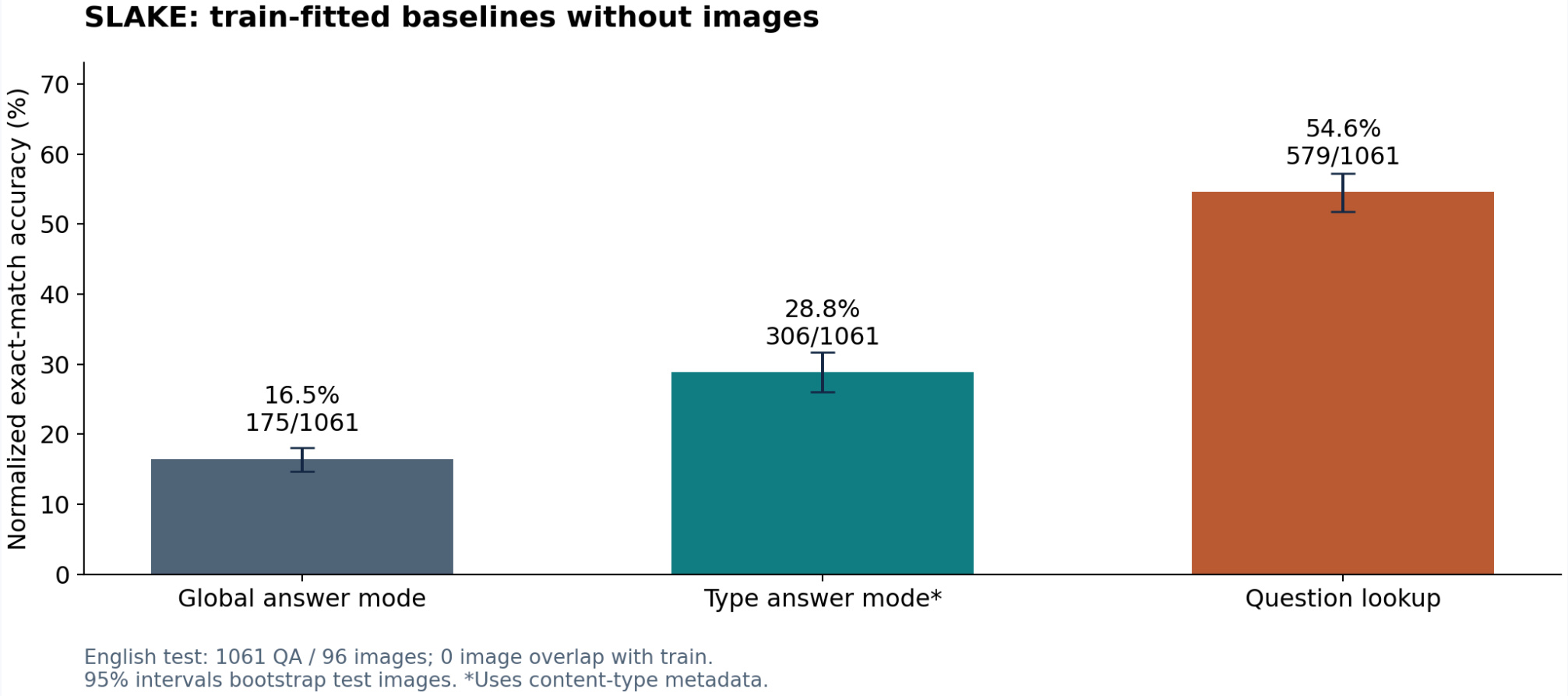


Original target includes:
“Our biopsy results confirm this diagnosis.”

MRI features ≠ evidence that a biopsy happened.

Keep the differential; add the pathology record before asserting confirmation.

Local benchmark scores have strong text baselines

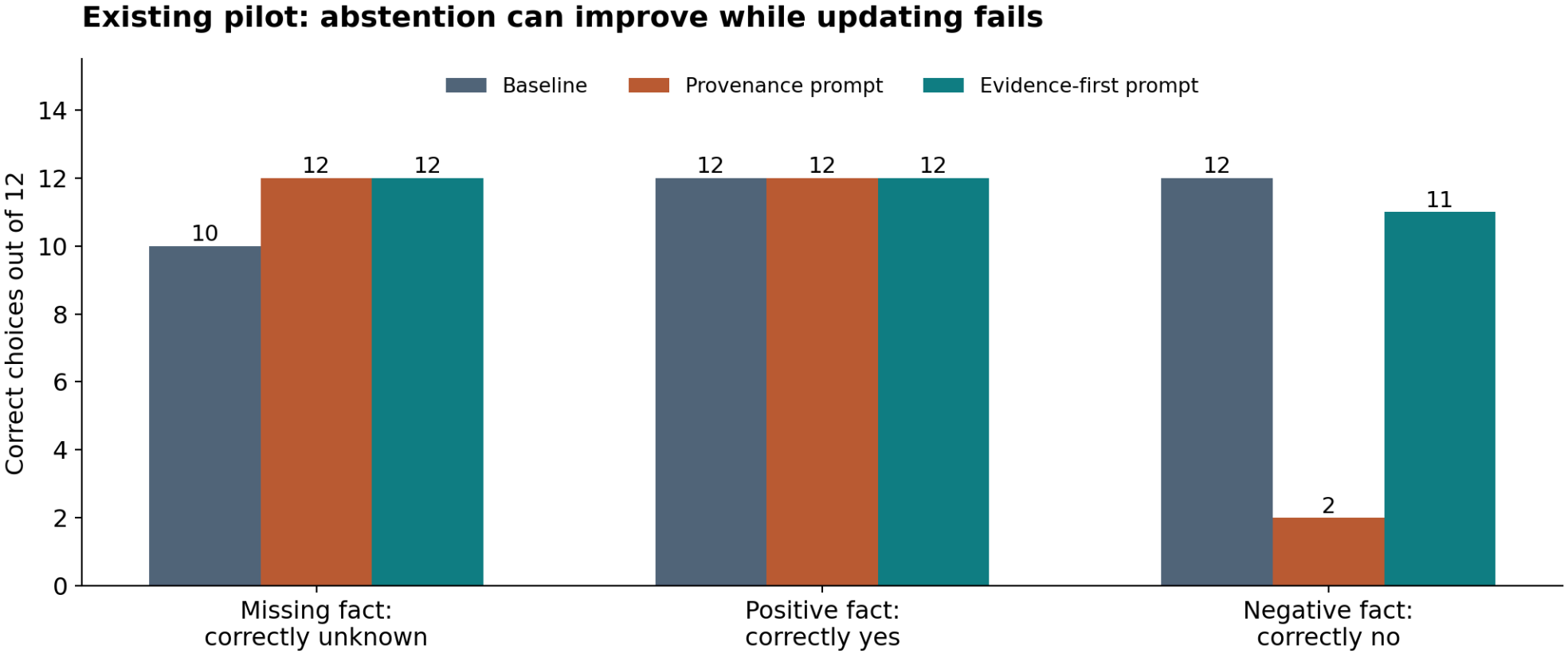


English test: 1061 QA / 96 images; 0 image overlap with train.
95% intervals bootstrap test images. *Uses content-type metadata.

Controls prevent an overbroad “language prior” story

- Some knowledge is intended: asking the function of an organ legitimately needs medical knowledge.
- A biopsy/IHC keyword may accompany an actual histology image. A keyword hit is not a missing-evidence label.
- PMC-VQA: predicting the train-mode option B gives 31.708% on test. Check answer-position balance.
- Visible dimensions can be answerable while longitudinal percentage change is not. Score atomic claims.

Existing pilot: abstention can conceal evidence suppression



Keep the decomposition; narrow what it proves

Prediction loss = intrinsic + missing context + lost encoding + predictor mismatch

- Population identity under explicit assumptions; not a per-case uncertainty label.
- Frozen-Z Fano bounds do not establish a universal pretraining/post-training boundary.
- Ablation likelihood gain is model dependence, not Shannon MI or expert evidence sufficiency.

A scoped experiment can establish the causal claim

- Annotate claims and required evidence on independent, patient/article-separated cases.
- Manipulate evidence availability: prior study, pathology report, missing panel. Keep questions fixed.
- Cross supervision alignment with a strong grounding objective at matched data and compute.
- Measure answerable-case accuracy, unsupported claims, correct updating and irrelevant-change stability together.

The primary outcome must require useful answers

Sufficient evidence

Answer adequately

Irrelevant change

Preserve supported claims

Required evidence absent

Qualify / request evidence

Valid evidence arrives

Update appropriately

What would weaken the hypothesis?

- Independent experts find little unavailable-evidence supervision in uniform samples.
- Aligned supervision fails to beat equal-size random cleaning or existing grounding objectives.
- Unsupported claims fall only because correct answerable cases are refused.
- A stronger frozen readout or better crop resolves the issue: investigate access/use before blaming irreversible encoding loss.

Advisor discussion: approve the testable core

Proceed with an evidence-aligned supervision study.

- Choose one modality and two evidence requirements.
- Secure independent clinical annotation and provenance checks.
- Agree on a matched causal ablation and a joint utility/grounding outcome.

The broad framework is reasonable. The decisive contribution still needs an experiment.

Sources: backbones and direct supervision precedents

P01 LLaVA-Med
https://arxiv.org/abs/2306.00890

P02 HuatuoGPT-Vision / PubMedVision
https://aclanthology.org/2024.emnlp-main.418/

P03 Lingshu / Lingshu-Med
https://arxiv.org/html/2506.07044v5

P04 MedGemma 1.5 Technical Report
https://arxiv.org/html/2604.05081v1

P05 Comparative evaluation of generative AI models for CXR report generation in the ED
https://pmc.ncbi.nlm.nih.gov/articles/PMC13574732/

P06 Removing hallucinated references to non-existent priors
https://proceedings.mlr.press/v193/ramesh22a.html

P07 DPO for suppressing hallucinated prior exams
https://arxiv.org/html/2406.06496v1

Sources: reasoning and benchmark evidence

P08 MedVLM-R1
https://papers.miccai.org/miccai-2025/0553-Paper3267.html

P09 Med-R1
https://arxiv.org/html/2503.13939v1

P10 GMAI-VL-R1
https://arxiv.org/html/2504.01886v1

P11 MedVLThinker
https://arxiv.org/html/2508.02669v1

P12 Beyond Accuracy: Evaluating Visual Grounding in Multimodal Medical Reasoning
https://arxiv.org/html/2603.03437v1

P13 MIRAGE: The Illusion of Visual Understanding
https://arxiv.org/html/2603.21687v2

P14 RadImageNet-VQA
https://arxiv.org/html/2512.17396v2

Sources: close methods

P15 Act as You Think / Multi-Consistency Reward
https://aclanthology.org/2026.acl-long.81/

P16 RadVLM with RL for reporting and grounding
https://arxiv.org/html/2512.10691v1

P17 ChestX-Reasoner: step-by-step verification from daily reports
https://www.nature.com/articles/s43856-026-01654-y

P18 MedReasoner
https://arxiv.org/html/2508.08177v1

P19 Med-OPD
https://arxiv.org/html/2607.16303v1

P20 MedTRACE
https://arxiv.org/html/2609.14823v1

Sources: attribution, abstention and calibration

P21 Spatial and factual RL for CXR reporting
https://arxiv.org/html/2609.31911v1

P22 CARE-MVLM
https://www.medrxiv.org/content/10.64898/2026.09.23.26363835v1.full

P23 RULE
https://aclanthology.org/2024.emnlp-main.62/

T01 V-usable information and pointwise V-information
https://proceedings.mlr.press/v162/ethayarajh22a.html

T02 Conformal Risk Control
https://arxiv.org/abs/2208.02814

T03 Language Models with Conformal Factuality Guarantees
https://proceedings.mlr.press/v235/mohri24a.html